$$\sin\left(90^\circ - r_a\right) = \frac{n_2}{n_1} \text{ (or) } \cos r_a = \frac{n_2}{n_1} \quad (6.40)$$

$$\sin r_a = \sqrt{1 - \cos^2 r_a}$$

Substituting for $\cos r_a$

$$\sin r_a = \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2} = \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}} \quad (6.41)$$

Substituting this in equation (6.37).

$$n_3 \sin i_a = n_1 \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}} = \sqrt{n_1^2 - n_2^2} \quad (6.42)$$

On further simplification,

$$\sin i_a = \frac{\sqrt{n_1^2 - n_2^2}}{n_3} \text{ (or) } \sin i_a = \sqrt{\frac{n_1^2 - n_2^2}{n_3^2}} \quad (6.43)$$

$$i_a = \sin^{-1}\left(\sqrt{\frac{n_1^2 - n_2^2}{n_3^2}}\right) \quad (6.44)$$

If outer medium is air, then $n_3 = 1$. The acceptance angle i_a becomes,

$$i_a = \sin^{-1}\left(\sqrt{n_1^2 - n_2^2}\right) \quad (6.45)$$

Light can have any angle of incidence from 0 to i_a with the normal at the end of the optical fibre forming a conical shape called ***acceptance cone*** as shown in Figure 6.29(b). In the equation (6.42), **the term $(n_3 \sin i_a)$ is called *numerical aperture NA*** of the optical fibre.

$$NA = n_3 \sin i_a = \sqrt{n_1^2 - n_2^2} \quad (6.46)$$

If outer medium is air, then $n_3 = 1$. The numerical aperture *NA* becomes,

$$NA = \sin i_a = \sqrt{n_1^2 - n_2^2} \quad (6.47)$$

EXAMPLE 6.10

A optical fibre is made up of a core material with refractive index 1.68 and a cladding material of refractive index 1.44. What is the acceptance angle of the fibre if it is kept in air medium without any cladding?

Solution

Given, $n_1 = 1.68$, $n_2 = 1.44$, $n_3 = 1$

Acceptance angle, $i_a = \sin^{-1}\left(\sqrt{n_1^2 - n_2^2}\right)$

$$i_a = \sin^{-1}\left(\sqrt{(1.68)^2 - (1.44)^2}\right) = \sin^{-1}(0.865)$$

$$i_a \approx 60^\circ$$

If there is no cladding then, $n_2 = 1$

Acceptance angle, $i_a = \sin^{-1}\left(\sqrt{n_1^2 - 1}\right)$

$$i_a = \sin^{-1}\left(\sqrt{(1.68)^2 - 1}\right) = \sin^{-1}(1.35)$$

$\sin^{-1}(\text{more than } 1)$ is not possible. But, this includes the range 0° to 90°. Hence, all the rays entering the core from flat surface will undergo total internal reflection.

Note: If there is no cladding then there is a condition on the refractive index (n_1) of the core.

$$i_a = \sin^{-1}\left(\sqrt{n_1^2 - 1}\right)$$

Here, as per mathematical rule, $\left(n_1^2 - 1\right) \le 1$ *or* $\left(n_1^2\right) \le 2$

or $n_1 \le \sqrt{2}$

Hence, in air (no cladding) the refractive index n_1 of the core should be, $n_1 \le 1.414$

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